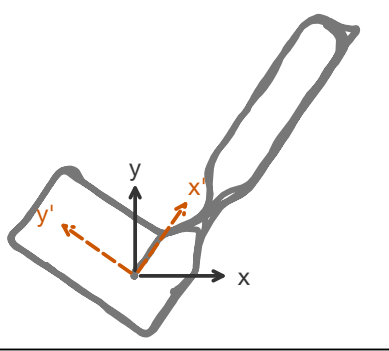
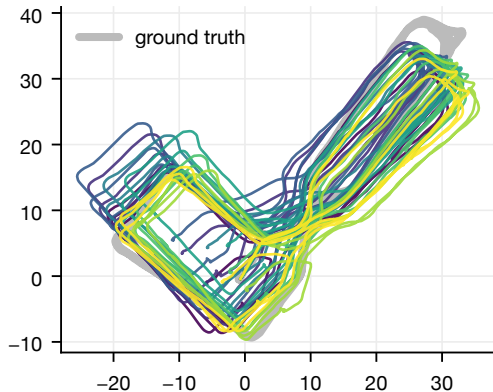


(a) representation



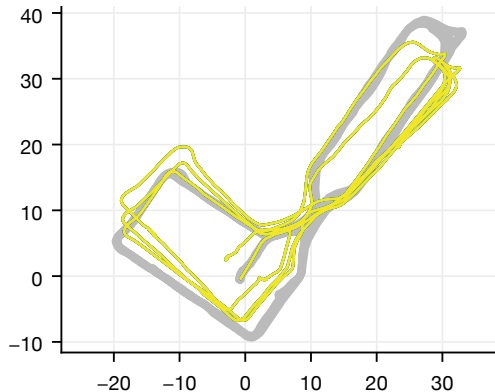
one motion, two horizontal frames
 X and $R_\psi X$ describe the same walk

(b) ordinary network (yaw-augmented)



recovered $R_{-\psi}F(R_\psi X)$
8 headings: ATE 2.546–4.915 m

(c) with HeadingNorm



recovered $R_{-\psi}F(R_\psi X)$
8 headings coincide: ATE 2.404 m